Chapter 7: Normalization

Database System Concepts, 7th Ed.

Overview of Normalization

Features of Good Relational Designs

- Suppose we combine *instructor* and *department* into *in_dep,* which represents the natural join on the relations *instructor* and *department*

ID	*name*	*salary*	*dept_name*	*building*	*budget*
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	Comp. Sci.	Taylor	100000
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

- There is repetition of information
- Need to use null values (if we add a new department with no instructors)

“好”的关系数据库设计

- 关系数据库是由一组关系构成的
- “好”的关系数据库设计要求我们找到一些“好”关系
- 包含“坏”关系的“坏”数据库设计具有某些问题

关系数据库中的问题

- 考虑以下数据库设计
  - 关系：worker (name, branch, manager)
  - 假设一个部门仅有一位经理。但反过来，一个经理可以是多个部门的经理。

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

关系数据库中的问题

- 该数据库设计有如下问题
  - 冗余
    - 思考：有几个元组存储了David是部门B经理的信息?
    - 同部门的每一个员工都不必要地重复关于谁是部门经理的信息

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

关系数据库中的问题

- 更新异常
  - 思考：如果部门B的经理变成Debbie，需要更新几个元组？
  - 如果一个部门的经理变动，我们必须更新部门里的每个员工以反映谁是新经理，否则就会出现这样的错误：该部门有两个经理

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

worker

name	branch	manager
Jones	A	Frank
Smith	B	Debbie
Curry	C	Frank
Larry	B	David
Julia	C	Frank

关系数据库中的问题

- 插入异常
  - 思考：假设成立一个新部门D。经理是kevin，没有员工。我们能增加关于部门D的信息么？
  - 如果一个部门还没有员工，我们就无法添加这个部门的信息

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank
null	D	Kevin

关系数据库中的问题

- 删除异常
  - 思考：在删除了部门B的所有员工以后，我们还能找出谁是部门B的经理么？
  - 如果一个部门里的所有员工都被删除了，谁是部门经理的信息也会被删除

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

worker

name	branch	manager
Jones	A	Frank
Curry	C	Frank
Julia	C	Frank

谁是部门B的经理?

什么导致了这些问题?

- 不良的数据依赖
  - 函数依赖(简记为FD)是一种数据依赖，它具有以下形式

$$\alpha \rightarrow \beta$$ (读作：α蕴涵β)

意义：当任意两个元组在属性集α上相等时，则它们在属性集β上也相等。即同一个α（的值），必然对应同一个β（的值）

什么导致了这些问题?

- 我们可以在worker关系中发现某些函数依赖，如下

  name → branch

  branch → manager

  name → manager

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

什么导致了这些问题?

- worker关系的这些函数依赖中, *branch → manager*是不良的
  - 由于左边的*branch*不是关系的码（这里指超码），同一*branch*可能出现多次，造成同一组合: *(branch, manager)*的重复

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank

如何解决这些问题?

- 分解
  - 把一个“坏” 关系分解为几个更小但是“好” 的关系
  - 例如

worker (name, branch, manager) →

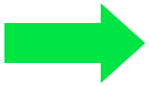
worker1 (name, branch),
branch (branch, manager)

如何解决这些问题?

- 在分解后消除了不良函数依赖，所以避免了同一组合的重复，解决了以上问题

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank



worker1

name	branch
Jones	A
Smith	B
Curry	C
Larry	B
Julia	C

branch

branch	manager
A	Frank
B	David
C	Frank

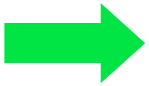
如何解决这些问题?

- 考虑另一种分解方案，如下所示

  worker (name, branch, manager) →
  worker2 (name, manager),
  branch (branch, manager)

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank



worker2

name	manager
Jones	Frank
Smith	David
Curry	Frank
Larry	David
Julia	Frank

Branch

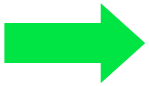
branch	manager
A	Frank
B	David
C	Frank

如何解决这些问题?

- 思考
  - 在分解后，你能找出Jones在哪个部门么?
- 不正确的分解可能带来新的问题：丢失某些信息
  - 例如，这里会丢失有关员工属于哪个部门的信息

worker

name	branch	manager
Jones	A	Frank
Smith	B	David
Curry	C	Frank
Larry	B	David
Julia	C	Frank



worker2

name	manager
Jones	Frank
Smith	David
Curry	Frank
Larry	David
Julia	Frank

Branch

branch	manager
A	Frank
B	David
C	Frank

思考与练习

- 观察以下关系，然后回答问题
  - 问题1：在这个关系上能找到哪些函数依赖？它们是“好”的还是“坏”的?
  - 问题2：它是一个“好”关系么？如果不是，存在哪些问题?

选修

学号	课程号	课程名	分数
01	A	数据库	90
01	B	C语言	85
02	A	数据库	70
02	C	算法分析	100
03	B	C语言	80

数据库设计理论

- 如何获得“好”的数据库设计?

  ① 判定其中的一个关系R是否“好”的，即有没有“坏”数据依赖

  ② 如果R不是好的，则要正确地分解为几个较小的好关系

  ③ 重复以上两步，直到全部关系都变成“好”的为止

Functional Dependencies

Functional Dependencies

- There are usually a variety of constraints (rules) on the data in the real world.
- For example, some of the constraints that are expected to hold in a university database are:
  - Students and instructors are uniquely identified by their ID.
  - Each student and instructor has only one name.
  - Each instructor and student is (primarily) associated with only one department.
  - Each department has only one value for its budget, and only one associated building.

Functional Dependencies (Cont.)

- An instance of a relation that satisfies all such real-world constraints is called a **legal instance** of the relation;
- A legal instance of a database is one where all the relation instances are legal instances
- Constraints on the set of legal relations.
- Require that the value for a certain set of attributes determines uniquely the value for another set of attributes.
- A functional dependency is a generalization of the notion of a *key.*

Functional Dependencies Definition

- Let R be a relation schema

$$\alpha \subseteq R \text{ and } \beta \subseteq R$$

- The **functional dependency**

$$\alpha \rightarrow \beta$$

holds on R if and only if for any legal relations $r(R)$, whenever any two tuples t_1 and t_2 of r agree on the attributes α, they also agree on the attributes β. That is,

$$t_1[\alpha] = t_2[\alpha] \Rightarrow t_1[\beta] = t_2[\beta]$$

- Example: Consider $r(A,B)$ with the following instance of r.

1	4
1	5
3	7

- On this instance, $B \rightarrow A$ hold; $A \rightarrow B$ does **NOT** hold,

补充：函数依赖

- 定义
  - R:　　一个关系
  - U:　　关系R中的所有属性
  - α, β : R中一或多个属性的集合

函数依赖

- 我们说关系R上存在以下函数依赖

$\alpha \rightarrow \beta$ (读作：α蕴涵β，或β依赖α)

的条件是<u>当且仅当</u>：任意两个元组如果在属性集α上相等，它们在属性集β上必然相等（同一个α对应同一个β）

$$t_1[\alpha] = t_2[\alpha] \Rightarrow t_1[\beta] = t_2[\beta]$$

函数依赖

- 例如
  - 考虑以下关系

student

name	college	city
Jones	SCNU	GZ
Smith	SCUT	GZ
Curry	TShingHua	BJ
Larry	PKU	BJ
Julia	PKU	BJ

  - 以上关系满足什么函数依赖?

Normal Forms

为什么使用范式?

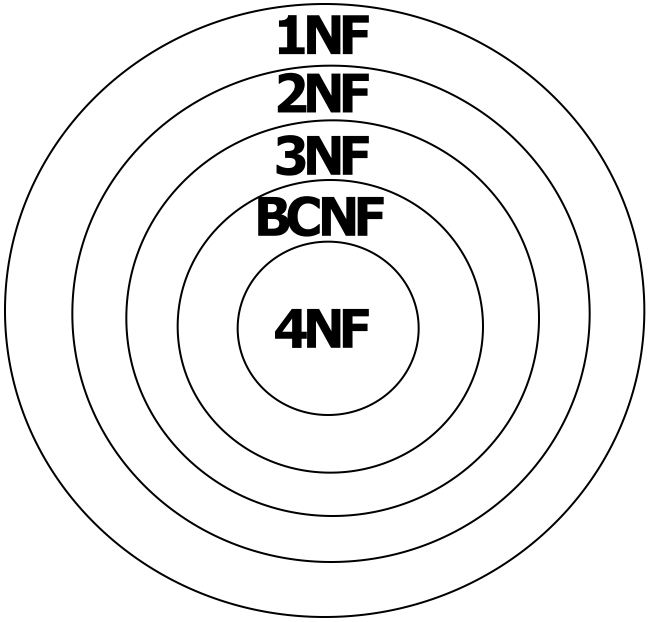
- “好”的关系数据库设计
  - 如何知道一个关系是否“好”，是否有“坏”的数据依赖
    - 判断属于哪一级别的范式
  - ……
- 高级范式与低级范式相比，是“更好”的关系，因为“坏”数据依赖更少

范式

- 范式
  - 满足一定要求的所有关系的全集
- 范式级别(从低级到高级)
  - 第一范式 (1NF)
  - 第二范式 (2NF)
  - 第三范式 (3NF)
  - BC范式 (BCNF)
  - 第四范式* (4NF)
  - ……
- 范式越高级，代表的关系就越“好”，要满足的要求也就越高

范式

- 不同范式之间的联系
  - 4NF ⊂ BCNF ⊂ 3NF ⊂ 2NF ⊂ 1NF
  - 高级范式是低级范式的子集
    - 范式越高，满足的要求也就越高；满足高要求的关系肯定能够满足低要求，所以高级范式中的关系肯定也在低级范式中



第一范式

- 要求
  - 关系的每个属性都是原子的
- 原子属性
  - 属性值不可再分
  - 例如. age, sex
- 非原子属性
  - 属性值可以再分，包括——
  - 复合属性，例如parents
  - 多值属性，例如phone-numbers

第一范式

- 例子
  - 下面关系中，哪些在1NF中，哪些不在?

Student

学号	姓名	课程号
S1	John	{C1，C2，C3}
S2	Smith	{C3，C5}

Student

学号	姓名	父母
S1	John	(Tom, Angie)
S2	Smith	(Mike, Sophie)

Student

学号	姓名	性别	年龄
S1	John	Male	22
S2	Smith	Male	23

一些术语和解释

- 例子关系
  - R(ABCDE)。候选码只有一个ABC，所以也是主码。
- 码属性（主属性, prime attribute）
  - 一个属性，出现在某个候选码（candidate key）中
  - 例如：A, B, C
- 非码属性（非主属性, non-prime attribute ）
  - 一个属性，不出现在任何一个候选码中
  - 例如：D, E
- （候选码）码的一部分
  - 就是候选码的真子集。
  - 例如，AB, BC, AC, A, B, C等等

一些术语和解释

- 超码
  - 具有唯一性的属性集。
  - 超码里面如果有多余的属性，那么去除后剩下的就是候选码。所以直观上看，超码含有或等于候选码。
  - 例如：ABC, ABCD, ABCE, ABCDE都是超码
- 非码
  - 非超码，也就是超码以外，不具有唯一性的属性集。
  - 直观上看，非码不含有候选码
  - 例如：AC, AD, CD, BD等都是非码

第二范式

- 要求
  - 关系在1NF中
  - 每个非码属性都完全函数依赖于候选码
    - 判断是否属于2NF：检查每个非码属性，是否可能依赖于候选码的一部分？有，不属于，否则属于。

第二范式

- 例子
  - 以下关系在1NF中？在2NF中？为什么？
  - 它们有我们以前讨论过的四个问题吗？

SC

学号	姓名	课程号	课程名
S1	John	C1	database
S2	Smith	C1	database
S1	John	C2	algorithm
S2	Smith	C2	algorithm

SC

学号	课程号	成绩
S1	C1	77
S2	C1	80
S1	C2	90
S2	C2	82

第三范式

- 要求
  - 关系在1NF中
  - 每个非码属性都非传递依赖于候选码
    - 判断是否属于3NF：检查每个非码属性，是否只依赖于超码（包含候选码）。如果有依赖于非超码，不属于，否则属于。
    - 注意，因为任何属性集都函数依赖于候选码(K→U)，所以非码属性A依赖于某个非码B的同时，也就传递依赖了候选码K (K→B→A)

第三范式

- 例子
  - 以下关系在3NF中？在2NF中？
  - 它们有我们以前讨论过的四个问题吗？

Student

学号	姓名	班号	班名
S1	小王	C1	会计班
S2	小张	C1	会计班
S3	小李	C2	电算班
S4	小孙	C2	电算班

Student

学号	姓名	班号
S1	小王	C1
S2	小张	C1
S3	小李	C2
S4	小孙	C2

Third Normal Form

- A relation schema R is in **third normal form (3NF)** if for all:

$$\alpha \rightarrow \beta \text{ in } F^+$$

at least one of the following holds:

  - $\alpha \rightarrow \beta$ is trivial (i.e., $\beta \in \alpha$)
  - α is a superkey for R
  - Each attribute A in $\beta - \alpha$ is contained in a candidate key for R.

(**NOTE**: each attribute may be in a different candidate key)

- If a relation is in BCNF it is in 3NF (since in BCNF one of the first two conditions above must hold).
- Third condition is a minimal relaxation of BCNF to ensure dependency preservation (will see why later).

3NF Example

- Consider a schema:

  dept_advisor(s_ID, i_ID, dept_name)
- With function dependencies:

  $i\_ID \rightarrow dept\_name$

  $s\_ID, dept\_name \rightarrow i\_ID$
- Two candidate keys = {*s_ID, dept_name*}, {*s_ID, i_ID* }
- We have seen before that *dept_advisor* is not in BCNF
- *R,* however, is in 3NF
  - *s_ID, dept_name* is a superkey
  - $i\_ID \rightarrow dept\_name$ *and* *i_ID* is NOT a superkey, but:
    - { *dept_name*} – {*i_ID* } = {*dept_name* } and
    - *dept_name* is contained in a candidate key

不用管主属性?

Redundancy in 3NF

- Consider the schema R below, which is in 3NF
  - $R = (J, K, L)$
  - $F = \{JK \rightarrow L, L \rightarrow K\}$
  - And an instance table:

J	L	K
j_1	l_1	k_1
j_2	l_1	k_1
j_3	l_1	k_1
null	l_2	k_2

- What is wrong with the table?
  - Repetition of information
  - Need to use null values (e.g., to represent the relationship l_2, k_2 where there is no corresponding value for J)

Boyce-Codd Normal Form

- A relation schema R is in BCNF with respect to a set F of functional dependencies if for all functional dependencies in F^+ of the form

$$\alpha \rightarrow \beta$$

where $\alpha \subseteq R$ and $\beta \subseteq R$, at least one of the following holds:

  - $\alpha \rightarrow \beta$ is trivial (i.e., $\beta \subseteq \alpha$)
  - α is a superkey for R

Boyce-Codd Normal Form (Cont.)

- Example schema that is ***not*** in BCNF:

  in_dep (*ID, name, salary, dept_name, building, budget*)

  because :

  - *dept_name*→ *building, budget*
    - holds on *in_dep*
    - but *dept_name* is not a superkey
- When decompose *in_dept* into *instructor* and *department*
  - *instructor* is in BCNF
  - *department* is in BCNF

例子：范式

- 例子
  - 考虑以下关系

R(S, T, C)

  - S: 学生;　T: 教师;　C: 课程

R

S	T	C
Smith	Jones	Java
Curry	Jones	Java
Smith	Frank	C++
Curry	Frank	C++
Larry	David	C++

范式

- 假设一个教师只教一门课程，但是一门课程有多个教师。也就是T→C
- 假设给定一个学生和一门课程，只有一个老师给他上这门课程。也就是SC→T

- 所以函数依赖集为：F={ T→C, SC→T }

R

S	T	C
Smith	Jones	Java
Curry	Jones	Java
Smith	Frank	C++
Curry	Frank	C++
Larry	David	C++

范式

$$R(S, T, C),\ \ F=\{\ T \rightarrow C,\ SC \rightarrow T\ \}$$

- 问题1: R的候选码是什么?
  - 候选码: ST, SC
  - 证明: $(ST)^+=(STC)$, $(SC)^+=(STC)$,所以ST, SC是超码. 而$(S)^+=(S)$, $(T)^+=(TC)$, $(C)^+=(C)$, 所以ST, SC的真子集都不是超码

R

S	T	C
Smith	Jones	Java
Curry	Jones	Java
Smith	Frank	C++
Curry	Frank	C++
Larry	David	C++

范式

R(S, T, C)，F={ T→C, SC→T },
候选码: ST, SC

- 问题2: R在3NF中么？在BCNF中么？
  - R在3NF中，因为R中没有非码属性
  - R不在BCNF中，因为T→C是非平凡的，且左边T不是超码

R

S	T	C
Smith	Jones	Java
Curry	Jones	Java
Smith	Frank	C++
Curry	Frank	C++
Larry	David	C++

BC范式

R(S, T, C)，F={ T→C, SC→T },
候选码: ST, SC

- 问题3: 它们有我们以前讨论过的四个问题吗?
  - 是的，因为组合(T, C)的值重复

R

S	T	C
Smith	Jones	Java
Curry	Jones	Java
Smith	Frank	C++
Curry	Frank	C++
Larry	David	C++

总结

检查每个函数依赖： $\alpha \rightarrow \beta$	结论
α 非码	非BCNF
α 非码， β 有非码属性	非3NF
α 是候选码的一部分，β有非码属性	非2NF

How good is BCNF?

- There are database schemas in BCNF that do not seem to be sufficiently normalized
- Consider a relation

  inst_info (ID, child_name, phone)

  - where an instructor may have more than one phone and can have multiple children
  - Instance of *inst_info*

ID	*child_name*	*phone*
99999	David	512-555-1234
99999	David	512-555-4321
99999	William	512-555-1234
99999	William	512-555-4321

How good is BCNF? (Cont.)

- There are no non-trivial functional dependencies and therefore the relation is in BCNF
- Insertion anomalies – i.e., if we add a phone 981-992-3443 to 99999, we need to add two tuples

  (99999, David, 981-992-3443)
  (99999, William, 981-992-3443)

Higher Normal Forms

- It is better to decompose *inst_info* into:
  - *inst_child*:

ID	*child_name*
99999	David
99999	William

  - *inst_phone:*

ID	*phone*
99999	512-555-1234
99999	512-555-4321

- This suggests the need for higher normal forms, such as Fourth Normal Form (4NF), which we shall see later

Decomposition

Decomposition

- The only way to avoid the repetition-of-information problem in the i*nst_dep* schema is to decompose it into two schemas – instructor and *department* schemas.
- Not all decompositions are good. Suppose we decompose

  employee(ID, name, street, city, salary)

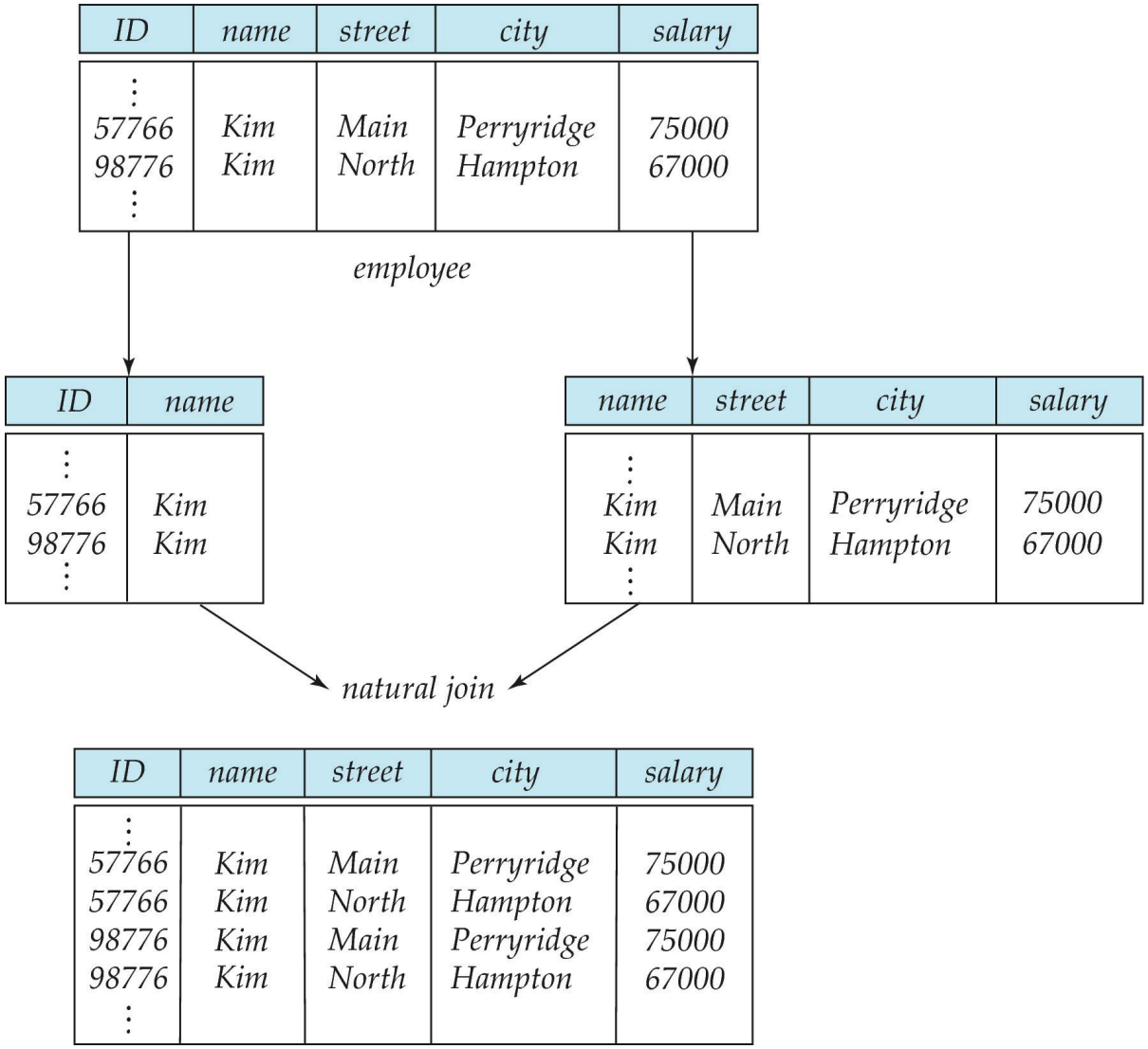
  into

  employee1 (*ID*, *name*)

  employee2 (*name*, *street, city, salary*)

  The problem arises when we have two employees with the same name
- The next slide shows how we lose information -- we cannot reconstruct the original *employee* relation -- and so, this is a **lossy decomposition**.

A Lossy Decomposition



Lossless Decomposition

- Let *R* be a relation schema and let R_1 and R_2 form a decomposition of R . That is R = R_1 U R_2
- We say that the decomposition is a **lossless decomposition** if there is no loss of information by replacing R with the two relation schemas R_1 U R_2
- Formally,

$$\prod_{R_1}(r) \bowtie \prod_{R_2}(r) = r$$

- And, conversely a decomposition is lossy if

$$r \subset \prod_{R_1}(r) \bowtie \prod_{R_2}(r) = r$$

Example of Lossless Decomposition

- Decomposition of $R = (A, B, C)$

  $R_1 = (A, B)$ $R_2 = (B, C)$

A	B	C
α	1	A
β	2	B

r

A	B
α	1
β	2

$\Pi_{A,B}(r)$

B	C
1	A
2	B

$\Pi_{B,C}(r)$

$\Pi_A (r) \bowtie \Pi_B (r)$

A	B	C
α	1	A
β	2	B

Normalization Theory

- Decide whether a particular relation *R* is in "good" form.
- In the case that a relation *R* is not in "good" form, decompose it into set of relations $\{R_1, R_2, ..., R_n\}$ such that
  - Each relation is in good form
  - The decomposition is a lossless decomposition
- Our theory is based on:
  - Functional dependencies
  - Multivalued dependencies

Closure of a Set of Functional Dependencies

- Given a set *F* set of functional dependencies, there are certain other functional dependencies that are logically implied by *F*.
  - If $A \rightarrow B$ and $B \rightarrow C$, then we can infer that $A \rightarrow C$
  - etc.
- The set of **all** functional dependencies logically implied by *F* is the **closure** of *F*.
- We denote the *closure* of *F* by F^+.

Keys and Functional Dependencies

- K is a superkey for relation schema R if and only if $K \rightarrow R$
- K is a candidate key for R if and only if
  - $K \rightarrow R$, and
  - for no $\alpha \subset K$, $\alpha \rightarrow R$
- Functional dependencies allow us to express constraints that cannot be expressed using superkeys. Consider the schema:

  in_dep (*ID, name, salary, dept_name, building, budget*).

  We expect these functional dependencies to hold:

  dept_name→ *building*

  ID → *building*

  but would not expect the following to hold:

  dept_name → *salary*

Use of Functional Dependencies

- We use functional dependencies to:
  - To test relations to see if they are legal under a given set of functional dependencies.
    - If a relation *r* is legal under a set *F* of functional dependencies, we say that *r* **satisfies** *F.*
  - To specify constraints on the set of legal relations
    - We say that *F* **holds on** *R* if all legal relations on *R* satisfy the set of functional dependencies *F.*
- Note: A specific instance of a relation schema may satisfy a functional dependency even if the functional dependency does not hold on all legal instances.
  - For example, a specific instance of *instructor* may, by chance, satisfy

    $name \rightarrow ID.$

Trivial Functional Dependencies

- *A* functional dependency is **trivial** if it is satisfied by all instances of a relation
- Example*:*
  - *ID, name* → *ID*
  - *name* → *name*
- In general, α → *β* is trivial if *β* ⊆ α
- 函数依赖α → β，当β ⊆ α 时是平凡的；否则是非平凡的
- 平凡的函数依赖总是成立的，为什么?
  - 换句话说，平凡的函数依赖是没有意义的，我们一般所讨论的函数依赖都应该排除这种情况。
-

如何让关系达到更高的范式?

- 分解
  - 把一个属于低级范式的“坏” 关系，分解为几个属于高级范式的 “好” 关系
  - 但是某些情况下分解会带来新的问题，比如信息丢失，这样的分解是不正确的。

- 要点

  什么样的分解方案才是正确的（不丢失信息的）？

  Dependency Preservation

  Lossless Decomposition

  (怎么做到？)

如何让关系达到更高的范式?

- 分解
  - 把一个属于低级范式的“坏” 关系，分解为几个属于高级范式的 “好” 关系
  - 但是某些情况下分解会带来新的问题，比如信息丢失，这样的分解是不正确的。

- 两个要点

  1. 什么样的分解方案才是正确的（不丢失信息的）？ —— Lossless-join decomposition and dependency preserving

  2. 怎么找到正确的分解方案？ —— 规范化

Decomposing a Schema into BCNF

- Let R be a schema R that is not in BCNF. Let $\alpha \rightarrow \beta$ be the FD that causes a violation of BCNF.
- We decompose R into:
  - $(\alpha \cup \beta)$
  - $(R - (\beta - \alpha))$
- In our example of *in_dep*,
  - α = *dept_name*
  - β = *building, budget*

  and *in_dep* is replaced by
  - $(\alpha \cup \beta)$ = (*dept_name, building, budget*)
  - $(R - (\beta - \alpha))$ = (*ID, name, dept_name, salary*)

Example

- $R = (A, B, C)$
  $F = \{A \rightarrow B, B \rightarrow C)$
- $R_1 = (A, B)$, $R_2 = (B, C)$
  - Lossless-join decomposition:

    $$R_1 \cap R_2 = \{B\} \quad \text{and } B \rightarrow BC$$

  - Dependency preserving
- $R_1 = (A, B)$, $R_2 = (A, C)$
  - Lossless-join decomposition:

    $$R_1 \cap R_2 = \{A\} \text{ and } A \rightarrow AB$$

  - Not dependency preserving
    (cannot check $B \rightarrow C$ without computing $R_1 \bowtie R_2$)

BCNF and Dependency Preservation

- It is not always possible to achieve both BCNF and dependency preservation
- Consider a schema:

  dept_advisor(s_ID, i_ID, department_name)
- With function dependencies:

  $i\_ID \rightarrow dept\_name$

  $s\_ID, dept\_name \rightarrow i\_ID$
- *dept_advisor* is not in BCNF
  - *i_ID* is not a superkey.
- Any decomposition of *dept_advisor* will not include all the attributes in

  $s\_ID, dept\_name \rightarrow i\_ID$
- Thus, the composition is NOT be dependency preserving

Comparison of BCNF and 3NF

- Advantages to 3NF over BCNF. It is always possible to obtain a 3NF design without sacrificing loss or dependency preservation.
- Disadvantages to 3NF.
  - We may have to use null values to represent some of the possible meaningful relationships among data items.
  - There is the problem of repetition of information.

Goals of Normalization

- Let R be a relation scheme with a set F of functional dependencies.
- Decide whether a relation scheme R is in "good" form.
- In the case that a relation scheme R is not in "good" form, need to decompose it into a set of relation scheme $\{R_1, R_2, ..., R_n\}$ such that:
  - Each relation scheme is in good form
  - The decomposition is a lossless decomposition
  - Preferably, the decomposition should be dependency preserving.

Functional-Dependency Theory

Functional-Dependency Theory Roadmap

- We now consider the formal theory that tells us which functional dependencies are implied logically by a given set of functional dependencies.
- We then develop algorithms to generate lossless decompositions into BCNF and 3NF
- We then develop algorithms to test if a decomposition is dependency-preserving

Closure of a Set of Functional Dependencies

- Given a set *F* set of functional dependencies, there are certain other functional dependencies that are logically implied by *F*.
  - If $A \rightarrow B$ and $B \rightarrow C$, then we can infer that $A \rightarrow C$
  - etc.
- The set of **all** functional dependencies logically implied by *F* is the **closure** of *F*.
- We denote the *closure* of *F* by F^+.
- *Why?*

Closure of a Set of Functional Dependencies

- We can compute F^+, the closure of F, by repeatedly applying **Armstrong's Axioms:**
  - **Reflexive rule:** if $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$
  - **Augmentation rule:** if $\alpha \rightarrow \beta$, then $\gamma\ \alpha \rightarrow \gamma\ \beta$
  - **Transitivity rule:** if $\alpha \rightarrow \beta$, and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$
- These rules are
  - **Sound** -- generate only functional dependencies that actually hold, and
  - **Complete** -- generate all functional dependencies that hold.

Example of F^+

- $R = (A, B, C, G, H, I)$
  $F = \{ A \rightarrow B$
  $A \rightarrow C$
  $CG \rightarrow H$
  $CG \rightarrow I$
  $B \rightarrow H\}$
- Some members of F^+
  - $A \rightarrow H$
    - by transitivity from $A \rightarrow B$ *and* $B \rightarrow H$
  - $AG \rightarrow I$
    - by augmenting $A \rightarrow C$ with G, to get $AG \rightarrow CG$ and then transitivity with $CG \rightarrow I$
  - $CG \rightarrow HI$
    - by augmenting $CG \rightarrow I$ to infer $CG \rightarrow CGI$,
      and augmenting of $CG \rightarrow H$ to infer $CGI \rightarrow HI$,
      and then transitivity

Closure of Functional Dependencies (Cont.)

- Additional rules:
  - **Union rule**: If $\alpha \rightarrow \beta$ holds *a*nd $\alpha \rightarrow \gamma$ holds, then $\alpha \rightarrow \beta\gamma$ holds.
  - **Decomposition rule**: If $\alpha \rightarrow \beta\gamma$ holds, then $\alpha \rightarrow \beta$ holds and $\alpha \rightarrow \gamma$ holds.
  - **Pseudotransitivity rule**:If $\alpha \rightarrow \beta$ holds *a*nd $\gamma\beta \rightarrow \delta$ holds, then $\alpha\gamma \rightarrow \delta$ holds**.**
- The above rules can be inferred from Armstrong's axioms.

Procedure for Computing F+

- To compute the closure of a set of functional dependencies F:

  $F^+ = F$
  repeat
    for each functional dependency f in F^+
      apply reflexivity and augmentation rules on f
      add the resulting functional dependencies to F^+
    for each pair of functional dependencies f_1 and f_2 in F^+
      if f_1 and f_2 can be combined using transitivity
        then add the resulting functional dependency to F^+
  until F^+ does not change any further

- **NOTE**: We shall see an alternative procedure for this task later

Closure of Attribute Sets

- Given a set of attributes α, define the ***closure*** of α **under** F (denoted by α^+) as the set of attributes that are functionally determined by α under F
- Algorithm to compute α^+, the closure of α under F

```
result := α;
while (changes to result) do
    for each β → γ in F do
      begin
        if β ⊆ result then  result := result ∪ γ
      end
```

Example of Attribute Set Closure

- $R = (A, B, C, G, H, I)$
- $F = \{A \rightarrow B$
  $A \rightarrow C$
  $CG \rightarrow H$
  $CG \rightarrow I$
  $B \rightarrow H\}$
- $(AG)^+$
  1. $result = AG$
  2. $result = ABCG$ $(A \rightarrow C$ and $A \rightarrow B)$
  3. $result = ABCGH$ $(CG \rightarrow H$ and $CG \subseteq AGBC)$
  4. $result = ABCGHI$ $(CG \rightarrow I$ and $CG \subseteq AGBCH)$
- Is AG a candidate key?
  1. Is AG a super key?
     1. Does $AG \rightarrow R$? == Is $R \supseteq (AG)^+$
  2. Is any subset of AG a superkey?
     1. Does $A \rightarrow R$? == Is $R \supseteq (A)^+$
     2. Does $G \rightarrow R$? == Is $R \supseteq (G)^+$
     3. In general: check for each subset of size $n-1$

Uses of Attribute Closure

There are several uses of the attribute closure algorithm:

- Testing for superkey:
  - To test if α is a superkey, we compute α^+, and check if α^+ contains all attributes of R.
- Testing functional dependencies
  - To check if a functional dependency $\alpha \rightarrow \beta$ holds (or, in other words, is in F^+), just check if $\beta \subseteq \alpha^+$.
  - That is, we compute α^+ by using attribute closure, and then check if it contains β.
  - Is a simple and cheap test, and very useful
- Computing closure of F
  - For each $\gamma \subseteq R$, we find the closure γ^+, and for each $S \subseteq \gamma^+$, we output a functional dependency $\gamma \rightarrow S$.

练习

- R (C, T, H, R, S)
- F = {C→T, HR→C, HT→R, HS→R}
- 问题1：求(HR) +?
- 问题2：HS是候选码么？
- 问题3：CH → S成立么？

Canonical Cover

- Suppose that we have a set of functional dependencies *F* on a relation schema. Whenever a user performs an update on the relation, the database system must ensure that the update does not violate any functional dependencies; that is, all the functional dependencies in *F* are satisfied in the new database state.
- If an update violates any functional dependencies in the set *F,* the system must roll back the update.
- We can reduce the effort spent in checking for violations by testing a simplified set of functional dependencies that has the same closure as the given set.
- This simplified set is termed the **canonical cover**
- To define canonical cover we must first define **extraneous attributes.**
  - An attribute of a functional dependency in *F* is **extraneous** if we can remove it without changing F^+

Extraneous Attributes

- Removing an attribute from the left side of a functional dependency could make it a stronger constraint.
  - For example, if we have $AB \rightarrow C$ and remove B, we get the possibly stronger result $A \rightarrow C$. It may be stronger because $A \rightarrow C$ logically implies $AB \rightarrow C$, but $AB \rightarrow C$ does not, on its own, logically imply $A \rightarrow C$
- But, depending on what our set F of functional dependencies happens to be, we may be able to remove B from $AB \rightarrow C$ safely.
  - For example, suppose that
  - $F = \{AB \rightarrow C, A \rightarrow D, D \rightarrow C\}$
  - Then we can show that F logically implies $A \rightarrow C$, making extraneous in $AB \rightarrow C$.

Extraneous Attributes (Cont.)

- Removing an attribute from the right side of a functional dependency could make it a weaker constraint.
  - For example, if we have AB $\rightarrow$ CD and remove C, we get the possibly weaker result AB $\rightarrow$ D. It may be weaker because using just AB $\rightarrow$ D, we can no longer infer AB $\rightarrow$ C.
- But, depending on what our set F of functional dependencies happens to be, we may be able to remove C from AB $\rightarrow$ CD safely.
  - For example, suppose that

    F = { AB $\rightarrow$ CD, A $\rightarrow$ C.
  - Then we can show that even after replacing AB $\rightarrow$ CD by AB $\rightarrow$ D, we can still infer $AB $\rightarrow$ C and thus AB $\rightarrow$ CD.

Extraneous Attributes

- An attribute of a functional dependency in F is **extraneous** if we can remove it without changing F^+
- Consider a set F of functional dependencies and the functional dependency $\alpha \rightarrow \beta$ in F.
  - **Remove from the left side**: Attribute A is **extraneous** in α if
    - $A \in \alpha$ and
    - F logically implies $(F - \{\alpha \rightarrow \beta\}) \cup \{(\alpha - A) \rightarrow \beta\}$.
  - **Remove from the right side**: Attribute A is **extraneous** in β if
    - $A \in \beta$ and
    - The set of functional dependencies

      $(F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - A)\}$ logically implies F.
- *Note:* implication in the opposite direction is trivial in each of the cases above, since a "stronger" functional dependency always implies a weaker one

Testing if an Attribute is Extraneous

- Let R be a relation schema and let F be a set of functional dependencies that hold on R. Consider an attribute in the functional dependency $\alpha \rightarrow \beta$.
- To test if attribute $A \in \beta$ is extraneous in β
  - Consider the set:
    $$F' = (F - \{\alpha \rightarrow \beta\}) \cup \{\alpha \rightarrow (\beta - A)\},$$
  - check that α^+ contains A; if it does, A is extraneous in β
- To test if attribute $A \in \alpha$ is extraneous in α
  - Let $\gamma = \alpha - \{A\}$. Check if $\gamma \rightarrow \beta$ can be inferred from F.
    - Compute γ^+ using the dependencies in F
    - If γ^+ includes all attributes in β then , A is extraneous in α

Examples of Extraneous Attributes

- Let $F = \{AB \rightarrow CD, A \rightarrow E, E \rightarrow C\}$
- To check if C is extraneous in $AB \rightarrow CD$, we:
  - Compute the attribute closure of AB under $F' = \{AB \rightarrow D, A \rightarrow E, E \rightarrow C\}$
  - The closure is $ABCDE$, which includes CD
  - This implies that C is extraneous

Canonical Cover

A **canonical cover** for F is a set of dependencies F_c such that

- F logically implies all dependencies in F_c , and
- F_c logically implies all dependencies in F, and
- No functional dependency in F_c contains an extraneous attribute, and
- Each left side of functional dependency in F_c is unique. That is, there are no two dependencies in F_c
  - $\alpha_1 \rightarrow \beta_1$ and $\alpha_2 \rightarrow \beta_2$ such that
  - $\alpha_1 = \alpha_2$

Canonical Cover

- To compute a canonical cover for *F*:

  repeat

  Use the union rule to replace any dependencies in *F* of the form

  $$\alpha_1 \rightarrow \beta_1 \text{ and } \alpha_1 \rightarrow \beta_2 \text{ with } \alpha_1 \rightarrow \beta_1 \beta_2$$

  Find a functional dependency $\alpha \rightarrow \beta$ in F_c with an extraneous attribute either in α or in β

  /* Note: test for extraneous attributes done using F_c, not F*/

  If an extraneous attribute is found, delete it from $\alpha \rightarrow \beta$

  until (F_c not change)

- Note: Union rule may become applicable after some extraneous attributes have been deleted, so it has to be re-applied

Example: Computing a Canonical Cover

- $R = (A, B, C)$
  $F = \{A \rightarrow BC$
  $B \rightarrow C$
  $A \rightarrow B$
  $AB \rightarrow C\}$
- Combine $A \rightarrow BC$ and $A \rightarrow B$ into $A \rightarrow BC$
  - Set is now $\{A \rightarrow BC, B \rightarrow C, AB \rightarrow C\}$
- A is extraneous in $AB \rightarrow C$
  - Check if the result of deleting A from $AB \rightarrow C$ is implied by the other dependencies
    - Yes: in fact, $B \rightarrow C$ is already present!
  - Set is now $\{A \rightarrow BC, B \rightarrow C\}$
- C is extraneous in $A \rightarrow BC$
  - Check if $A \rightarrow C$ is logically implied by $A \rightarrow B$ and the other dependencies
    - Yes*:* using transitivity on $A \rightarrow B$ *and* $B \rightarrow$ C.
      - Can use attribute closure of A in more complex cases
- The canonical cover is: $A \rightarrow B$
  $B \rightarrow C$

Algorithm for Decomposition Using Functional Dependencies

如何让关系达到更高的范式?

- 分解
  - 把一个属于低级范式的“坏” 关系，分解为几个属于高级范式的 “好” 关系
  - 但是某些情况下分解会带来新的问题，比如信息丢失，这样的分解是不正确的。

- 两个要点
  1. 什么样的分解方案才是正确的（不丢失信息的）？ —— 无损连接分解：
  2. Dependency Preservation
  3. Lossless Decomposition

Dependency Preservation

- Let F_i be the set of dependencies F^+ that include only attributes in R_i.
  - A decomposition is **dependency preserving**, if

$$(F_1 \cup F_2 \cup \ldots \cup F_n)^+ = F^+$$

- Using the above definition, testing for dependency preservation take exponential time.
- Not that if a decomposition is NOT dependency preserving then checking updates for violation of functional dependencies may require computing joins, which is expensive.

Dependency Preservation (Cont.)

- Let F be the set of dependencies on schema R and let $R_1, R_2, .., R_n$ be a decomposition of R.
- The restriction of F to R_i is the set F_i of all functional dependencies in F^+ that include **only** attributes of R_i.
- Since all functional dependencies in a restriction involve attributes of only one relation schema, it is possible to test such a dependency for satisfaction by checking only one relation.
- Note that the definition of restriction uses all dependencies in in F^+, not just those in F.
- The set of restrictions $F_1, F_2, .., F_n$ is the set of functional dependencies that can be checked efficiently.

Testing for Dependency Preservation

- To check if a dependency $\alpha \rightarrow \beta$ is preserved in a decomposition of R into $R_1, R_2, \ldots, R_n$, we apply the following test (with attribute closure done with respect to F)
  - *result* = α
    repeat
    for each R_i in the decomposition
    $t = (result \cap R_i)^+ \cap R_i$
    $result = result \cup t$
    until (*result* does not change)
  - If *result* contains all attributes in β, then the functional dependency $\alpha \rightarrow \beta$ is preserved.
- We apply the test on all dependencies in F to check if a decomposition is dependency preserving
- This procedure takes polynomial time, instead of the exponential time required to compute F^+ and $(F_1 \cup F_2 \cup \ldots \cup F_n)^+$

Example

- $R = (A, B, C)$
  $F = \{A \rightarrow B$
  $B \rightarrow C\}$
  Key = $\{A\}$
- R is not in BCNF
- Decomposition $R_1 = (A, B)$, $R_2 = (B, C)$
  - R_1 and R_2 in BCNF
  - Lossless-join decomposition
  - Dependency preserving

Lossless Decomposition

- We can use functional dependencies to show when certain decomposition are lossless.
- For the case of $R = (R_1, R_2)$, we require that for all possible relations r on schema R

$$r = \prod_{R1}(r) \bowtie \prod_{R2}(r)$$

- A decomposition of R into R_1 and R_2 is lossless decomposition if at least one of the following dependencies is in F^+:
  - $R_1 \cap R_2 \rightarrow R_1$
  - $R_1 \cap R_2 \rightarrow R_2$
- The above functional dependencies are a sufficient condition for lossless join decomposition; the dependencies are a necessary condition only if all constraints are functional dependencies

Example

- $R = (A, B, C)$
  $F = \{A \rightarrow B, B \rightarrow C)$
- $R_1 = (A, B), \quad R_2 = (B, C)$
  - Lossless decomposition:

    $$R_1 \cap R_2 = \{B\} \text{ and } B \rightarrow BC$$
- $R_1 = (A, B), \quad R_2 = (A, C)$
  - Lossless decomposition:

    $$R_1 \cap R_2 = \{A\} \text{ and } A \rightarrow AB$$
- *Note:*
  - $B \rightarrow BC$

    is a shorthand notation for
  - $B \rightarrow \{B, C\}$

Example

- *R(C, T, H, R, S)*
  F = { C→T, HR→C, HT→R, HS→R }

- 问题1: *R* → *R1(C, H, S)*, *R2(C, T, H, R)*. 这一分解是无损的么?

Dependency Preservation

- Testing functional dependency constraints each time the database is updated can be costly
- It is useful to design the database in a way that constraints can be tested efficiently.
- If testing a functional dependency can be done by considering just one relation, then the cost of testing this constraint is low
- When decomposing a relation it is possible that it is no longer possible to do the testing without having to perform a Cartesian Produced.
- A decomposition that makes it computationally hard to enforce functional dependency is said to be NOT **dependency preserving**.

Dependency Preservation Example

- Consider a schema:

  dept_advisor(s_ID, i_ID, department_name)

- With function dependencies:

  $i\_ID \rightarrow dept\_name$

  $s\_ID, dept\_name \rightarrow i\_ID$

- In the above design we are forced to repeat the department name once for each time an instructor participates in a *dept_advisor* relationship.
- To fix this, we need to decompose *dept_advisor*
- Any decomposition will not include all the attributes in

  $s\_ID, dept\_name \rightarrow i\_ID$

- Thus, the composition NOT be dependency preserving

Testing for BCNF

- To check if a non-trivial dependency $\alpha \rightarrow \beta$ causes a violation of BCNF
  1. compute α^+ (the attribute closure of α), and
  2. verify that it includes all attributes of R, that is, it is a superkey of R.
- **Simplified test**: To check if a relation schema R is in BCNF, it suffices to check only the dependencies in the given set F for violation of BCNF, rather than checking all dependencies in F^+.
  - If none of the dependencies in F causes a violation of BCNF, then none of the dependencies in F^+ will cause a violation of BCNF either.
- However, **simplified test** using only F is incorrect when testing a relation in a decomposition of R
  - Consider $R = (A, B, C, D, E)$, with $F = \{ A \rightarrow B, BC \rightarrow D\}$
    - Decompose R into $R_1 = (A,B)$ and $R_2 = (A,C,D, E)$
    - Neither of the dependencies in F contain only attributes from (A,C,D,E) so we might be mislead into thinking R_2 satisfies BCNF.
    - In fact, dependency $AC \rightarrow D$ in F^+ shows R_2 is not in BCNF.

Testing Decomposition for BCNF

To check if a relation R_i in a decomposition of R is in BCNF

- Either test R_i for BCNF with respect to the **restriction** of F^+ to R_i (that is, all FDs in F^+ that contain only attributes from R_i)
- Or use the original set of dependencies F that hold on R, but with the following test:
  - for every set of attributes $\alpha \subseteq R_i$, check that α^+ (the attribute closure of α) either includes no attribute of $R_i - \alpha$, or includes all attributes of R_i.
  - If the condition is violated by some $\alpha \rightarrow \beta$ in F^+, the dependency
    $$\alpha \rightarrow (\alpha^+ - \alpha) \cap R_i$$
    can be shown to hold on R_i, and R_i violates BCNF.
  - We use above dependency to decompose R_i

BCNF Decomposition Algorithm

result := {*R* };
done := false;
compute F^+;
while (not *done)* **do**
  if (there is a schema R_i in *result* that is not in BCNF)
    then begin
      let $\alpha \rightarrow \beta$ be a nontrivial functional dependency that
        holds on R_i such that $\alpha \rightarrow R_i$ is not in F^+,
        and $\alpha \cap \beta = \varnothing$;
      result := $\{ (result - R_i) \cup (R_i - \beta) \} \cup \{(\alpha, \beta)\}$;
      end
    else *done* := **true;**

Note: each R_i is in BCNF, and decomposition is lossless-join.

Example of BCNF Decomposition

- *class* (*course_id*, *title*, *dept_name*, *credits*, *sec_id*, *semester*, *year*, *building*, *room_number*, *capacity*, *time_slot_id*)
- Functional dependencies:
  - *course_id*→ *title*, *dept_name*, *credits*
  - *building*, *room_number*→*capacity*
  - *course_id*, *sec_id*, *semester*, *year*→*building*, *room_number*, *time_slot_id*
- A candidate key {*course_id*, *sec_id*, *semester*, *year*}.
- BCNF Decomposition:
  - *course_id*→ *title*, *dept_name*, *credits* holds
    - but *course_id* is not a superkey.
  - We replace *class* by:
    - *course*(*course_id*, *title*, *dept_name*, *credits*)
    - *class-1* (*course_id*, *sec_id*, *semester*, *year*, *building*, *room_number, capacity*, *time_slot_id*)

BCNF Decomposition (Cont.)

- *course* is in BCNF
  - How do we know this?
- *building*, *room_number*→*capacity* holds on *class-1*
  - but {*building*, *room_number*} is not a superkey for *class-1*.
  - We replace *class-1* by:
    - *classroom* (*building*, *room_number*, *capacity*)
    - *section* (*course_id*, *sec_id*, *semester*, *year*, *building*, *room_number*, *time_slot_id*)
- *classroom* and *section* are in BCNF.

Third Normal Form

- There are some situations where
  - BCNF is not dependency preserving, and
  - efficient checking for FD violation on updates is important
- Solution: define a weaker normal form, called Third Normal Form (3NF)
  - Allows some redundancy (with resultant problems; we will see examples later)
  - But functional dependencies can be checked on individual relations without computing a join.
  - There is always a lossless-join, dependency-preserving decomposition into 3NF.

3NF Example -- Relation *dept_advisor*

- *dept_advisor* (*s_ID, i_ID, dept_name*)
  $F = \{s\_ID, dept\_name \rightarrow i\_ID,\ i\_ID \rightarrow dept\_name\}$
- Two candidate keys: *s_ID, dept_name,* and *i_ID, s_ID*
- *R* is in 3NF
  - *s_ID, dept_name* $\rightarrow$ *i_ID* *s_ID*
    - *dept_name* is a superkey
  - *i_ID* $\rightarrow$ *dept_name*
    - *dept_name* is contained in a candidate key

Testing for 3NF

- Need to check only FDs in F, need not check all FDs in F^+.
- Use attribute closure to check for each dependency $\alpha \rightarrow \beta$, if α is a superkey.
- If α is not a superkey, we have to verify if each attribute in β is contained in a candidate key of R
  - This test is rather more expensive, since it involve finding candidate keys
  - Testing for 3NF has been shown to be NP-hard
  - Interestingly, decomposition into third normal form (described shortly) can be done in polynomial time

3NF Decomposition Algorithm

Let F_c be a canonical cover for F;
$i := 0$;
for each functional dependency $\alpha \rightarrow \beta$ in F_c **do**
  if none of the schemas R_j, $1 \le j \le i$ contains $\alpha \beta$
    then begin
      $i := i + 1$;
      $R_i := \alpha \beta$
    end
if none of the schemas R_j, $1 \le j \le i$ contains a candidate key for R
  then begin
    $i := i + 1$;
    R_i := any candidate key for R;
  end
/* Optionally, remove redundant relations */
repeat
if any schema R_j is contained in another schema R_k
  then /* delete R_j */
    $R_j = R$;;
    $i=i-1$;
return $(R_1, R_2, ..., R_i)$

3NF Decomposition Algorithm (Cont.)

Above algorithm ensures

- Each relation schema R_i is in 3NF
- Decomposition is dependency preserving and lossless-join
- Proof of correctness is at end of this presentation (click here)

3NF Decomposition: An Example

- Relation schema:

  cust_banker_branch = (*customer_id, employee_id, branch_name, type*)

- The functional dependencies for this relation schema are:
  - *customer_id, employee_id* → *branch_name, type*
  - *employee_id* → *branch_name*
  - *customer_id, branch_name* → *employee_id*
- We first compute a canonical cover
  - *branch_name* is extraneous in the r.h.s. of the 1st dependency
  - No other attribute is extraneous, so we get F_C =

    customer_id, employee_id → *type*
    employee_id → *branch_name*
    customer_id, branch_name → *employee_id*

3NF Decompsition Example (Cont.)

- The **for** loop generates following 3NF schema:

  (*customer_id, employee_id, type*)

  (*employee_id, branch_name*)

  (*customer_id, branch_name, employee_id)*

  - Observe that (*customer_id, employee_id, type*) contains a candidate key of the original schema, so no further relation schema needs be added
- At end of for loop, detect and delete schemas, such as (*employee_id, branch_name*), which are subsets of other schemas
  - result will not depend on the order in which FDs are considered
- The resultant simplified 3NF schema is:

  (*customer_id, employee_id, type*)

  (*customer_id, branch_name, employee_id)*

规范化到3NF

- 例
  - 输入: R(A, B, C, D, E), F = (A→B, C→D, D → E)
  - 候选码: AC

1. R1(AB)
2. R1(AB), R2(CD)
3. R1(AB), R2(CD), R3(DE)
4. R1(AB), R2(CD), R3(DE), R4(AC)

  - Output: R1(AB), R2(CD), R3(DE), R4(AC)

Comparison of BCNF and 3NF

- It is always possible to decompose a relation into a set of relations that are in 3NF such that:
  - The decomposition is lossless
  - The dependencies are preserved
- It is always possible to decompose a relation into a set of relations that are in BCNF such that:
  - The decomposition is lossless
  - It may not be possible to preserve dependencies.

Design Goals

- Goal for a relational database design is:
  - BCNF.
  - Lossless join.
  - Dependency preservation.
- If we cannot achieve this, we accept one of
  - Lack of dependency preservation
  - Redundancy due to use of 3NF
- Interestingly, SQL does not provide a direct way of specifying functional dependencies other than superkeys.

  Can specify FDs using assertions, but they are expensive to test, (and currently not supported by any of the widely used databases!)
- Even if we had a dependency preserving decomposition, using SQL we would not be able to efficiently test a functional dependency whose left hand side is not a key.

Multivalued Dependencies

Multivalued Dependencies (MVDs)

- Suppose we record names of children, and phone numbers for instructors:
  - *inst_child*(*ID*, *child_name*)
  - *inst_phone*(*ID*, *phone_number*)
- If we were to combine these schemas to get
  - *inst_info*(*ID*, *child_name*, *phone_number*)
  - Example data:
    (99999, David, 512-555-1234)
    (99999, David, 512-555-4321)
    (99999, William, 512-555-1234)
    (99999, William, 512-555-4321)
- This relation is in BCNF
  - Why?

Multivalued Dependencies

- Let R be a relation schema and let $\alpha \subseteq R$ and $\beta \subseteq R$. The **multivalued dependency**

$$\alpha \twoheadrightarrow \beta$$

holds on R if in any legal relation $r(R)$, for all pairs for tuples t_1 and t_2 in r such that $t_1[\alpha] = t_2[\alpha]$, there exist tuples t_3 and t_4 in r such that:

$$t_1[\alpha] = t_2[\alpha] = t_3[\alpha] = t_4[\alpha]$$
$$t_3[\beta] = t_1[\beta]$$
$$t_3[R - \beta] = t_2[R - \beta]$$
$$t_4[\beta] = t_2[\beta]$$
$$t_4[R - \beta] = t_1[R - \beta]$$

MVD -- Tabular representation

- Tabular representation of $\alpha \rightarrow\rightarrow \beta$

	α	β	$R - \alpha - \beta$
t_1	$a_1 \ldots a_i$	$a_{i+1} \ldots a_j$	$a_{j+1} \ldots a_n$
t_2	$a_1 \ldots a_i$	$b_{i+1} \ldots b_j$	$b_{j+1} \ldots b_n$
t_3	$a_1 \ldots a_i$	$a_{i+1} \ldots a_j$	$b_{j+1} \ldots b_n$
t_4	$a_1 \ldots a_i$	$b_{i+1} \ldots b_j$	$a_{j+1} \ldots a_n$

MVD (Cont.)

- Let R be a relation schema with a set of attributes that are partitioned into 3 nonempty subsets.

  $$Y, Z, W$$

- We say that $Y \twoheadrightarrow Z$ (Y **multidetermines** Z) if and only if for all possible relations $r(R)$

  $$< y_1, z_1, w_1 > \in r \text{ and } < y_1, z_2, w_2 > \in r$$

  then

  $$< y_1, z_1, w_2 > \in r \text{ and } < y_1, z_2, w_1 > \in r$$

- Note that since the behavior of Z and W are identical it follows that

  $Y \twoheadrightarrow Z$ if $Y \twoheadrightarrow W$

Example

- In our example:

$$ID \twoheadrightarrow child\_name$$
$$ID \twoheadrightarrow phone\_number$$

- The above formal definition is supposed to formalize the notion that given a particular value of Y (ID) it has associated with it a set of values of Z *(child_name)* and a set of values of W *(phone_number)*, and these two sets are in some sense independent of each other.
- Note:
  - If $Y \rightarrow Z$ then $Y \twoheadrightarrow Z$
  - Indeed we have (in above notation) $Z_1 = Z_2$
    The claim follows.

Use of Multivalued Dependencies

- We use multivalued dependencies in two ways:
  1. To test relations to **determine** whether they are legal under a given set of functional and multivalued dependencies
  2. To specify **constraints** on the set of legal relations. We shall concern ourselves *only* with relations that satisfy a given set of functional and multivalued dependencies.
- If a relation *r* fails to satisfy a given multivalued dependency, we can construct a relations *r′* that does satisfy the multivalued dependency by adding tuples to *r.*

Theory of MVDs

- From the definition of multivalued dependency, we can derive the following rule:
  - If $\alpha \rightarrow \beta$, then $\alpha \twoheadrightarrow \beta$

  That is, every functional dependency is also a multivalued dependency
- The **closure** D^+ of *D* is the set of all functional and multivalued dependencies logically implied by *D*.
  - We can compute D^+ from *D*, using the formal definitions of functional dependencies and multivalued dependencies.
  - We can manage with such reasoning for very simple multivalued dependencies, which seem to be most common in practice
  - For complex dependencies, it is better to reason about sets of dependencies using a system of inference rules (Appendix C).

Fourth Normal Form

- A relation schema R is in **4NF** with respect to a set D of functional and multivalued dependencies if for all multivalued dependencies in D^+ of the form $\alpha \rightarrow\!\!\!\rightarrow \beta$, where $\alpha \subseteq R$ and $\beta \subseteq R$, at least one of the following hold:
  - $\alpha \rightarrow\!\!\!\rightarrow \beta$ is trivial (i.e., $\beta \subseteq \alpha$ or $\alpha \cup \beta = R$)
  - α is a superkey for schema R
- If a relation is in 4NF it is in BCNF

Restriction of Multivalued Dependencies

- The restriction of D to R_i is the set D_i consisting of
  - All functional dependencies in D^+ that include only attributes of R_i
  - All multivalued dependencies of the form

    $\alpha \rightarrow\rightarrow (\beta \cap R_i)$

    where $\alpha \subseteq R_i$ and $\alpha \rightarrow\rightarrow \beta$ is in D^+

4NF Decomposition Algorithm

result: = {*R*};
done := false;
compute D^+;
Let D_i denote the restriction of D^+ to R_i

while (**not** *done*)
  if (there is a schema $\mathbf{R}_i$ in *result* that is not in 4NF) **then**
    begin

      let $\alpha \rightarrow\!\!\!\rightarrow \beta$ be a nontrivial multivalued dependency that holds
        on R_i such that $\alpha \rightarrow R_i$ is not in D_i, and $\alpha \cap \beta = \phi$;
      result := (*result* - R_i) $\cup$ (R_i - β) $\cup$ (α, β);
    end
  else *done*:= true;

Note: each R_i is in 4NF, and decomposition is lossless-join

Example

- $R = (A, B, C, G, H, I)$

  $F = \{ A \twoheadrightarrow B$

  $B \twoheadrightarrow HI$

  $CG \twoheadrightarrow H \}$

- R is not in 4NF since $A \twoheadrightarrow B$ and A is not a superkey for R
- Decomposition

  a) $R_1 = (A, B)$ (R_1 is in 4NF)

  b) $R_2 = (A, C, G, H, I)$ (R_2 is not in 4NF, decompose into R_3 and R_4)

  c) $R_3 = (C, G, H)$ (R_3 is in 4NF)

  d) $R_4 = (A, C, G, I)$ (R_4 is not in 4NF, decompose into R_5 and R_6)

  - $A \twoheadrightarrow B$ and $B \twoheadrightarrow HI$ ➔ $A \twoheadrightarrow HI$, (MVD transitivity), and
  - and hence $A \twoheadrightarrow I$ *(MVD restriction to R_4)*

  e) $R_5 = (A, I)$ (R_5 is in 4NF)

  f) $R_6 = (A, C, G)$ (R_6 is in 4NF)

Additional issues

Further Normal Forms

- **Join dependencies** generalize multivalued dependencies
  - lead to **project-join normal form (PJNF)** (also called **fifth normal form**)
- A class of even more general constraints, leads to a normal form called **domain-key normal form**.
- Problem with these generalized constraints: are hard to reason with, and no set of sound and complete set of inference rules exists.
- Hence rarely used

Overall Database Design Process

We have assumed schema *R* is given

- *R* could have been generated when converting E-R diagram to a set of tables.
- *R* could have been a single relation containing *all* attributes that are of interest (called **universal relation**).
- Normalization breaks *R* into smaller relations.
- *R* could have been the result of some ad hoc design of relations, which we then test/convert to normal form.

ER Model and Normalization

- When an E-R diagram is carefully designed, identifying all entities correctly, the tables generated from the E-R diagram should not need further normalization.
- However, in a real (imperfect) design, there can be functional dependencies from non-key attributes of an entity to other attributes of the entity
  - Example: an *employee* entity with
    - attributes
      department_name and *building*,
    - functional dependency
      department_name $\rightarrow$ *building*
    - Good design would have made department an entity
- Functional dependencies from non-key attributes of a relationship set possible, but rare --- most relationships are binary

Denormalization for Performance

- May want to use non-normalized schema for performance
- For example, displaying *prereqs* along with *course_id,* and *title* requires join of *course* with *prereq*
- Alternative 1: Use denormalized relation containing attributes of *course* as well as *prereq* with all above attributes
  - faster lookup
  - extra space and extra execution time for updates
  - extra coding work for programmer and possibility of error in extra code
- Alternative 2: use a materialized view defined a *course* ⋈ *prereq*
  - Benefits and drawbacks same as above, except no extra coding work for programmer and avoids possible errors

Other Design Issues

- Some aspects of database design are not caught by normalization
- Examples of bad database design, to be avoided:

  Instead of *earnings* (*company_id, year, amount*), use
  - *earnings_2004, earnings_2005, earnings_2006*, etc., all on the schema (*company_id, earnings*).
    - Above are in BCNF, but make querying across years difficult and needs new table each year
  - *company_year* (*company_id, earnings_2004, earnings_2005, earnings_2006*)
    - Also in BCNF, but also makes querying across years difficult and requires new attribute each year.
    - Is an example of a **crosstab**, where values for one attribute become column names
    - Used in spreadsheets, and in data analysis tools

Modeling Temporal Data

- **Temporal data** have an association time interval during which the data are *valid.*
- A **snapshot** is the value of the data at a particular point in time
- Several proposals to extend ER model by adding valid time to
  - attributes, e.g., address of an instructor at different points in time
  - entities, e.g., time duration when a student entity exists
  - relationships, e.g., time during which an instructor was associated with a student as an advisor.
- But no accepted standard
- Adding a temporal component results in functional dependencies like

  $ID \rightarrow street, city$

  not holding, because the address varies over time
- A **temporal functional dependency** $X \rightarrow Y$ holds on schema R if the functional dependency $X \rightarrow Y$ holds on all snapshots for all legal instances r (R).

Modeling Temporal Data (Cont.)

- In practice, database designers may add start and end time attributes to relations
  - E.g., *course*(*course_id, course_title*) is replaced by

    course(*course_id, course_title, start, end*)
  - Constraint: no two tuples can have overlapping valid times
    - Hard to enforce efficiently
- Foreign key references may be to current version of data, or to data at a point in time
  - E.g., student transcript should refer to course information at the time the course was taken

End of Chapter 7

Proof of Correctness of 3NF Decomposition Algorithm

Correctness of 3NF Decomposition Algorithm

- 3NF decomposition algorithm is dependency preserving (since there is a relation for every FD in F_c)
- Decomposition is lossless
  - A candidate key (C) is in one of the relations R_i in decomposition
  - Closure of candidate key under F_c must contain all attributes in R.
  - Follow the steps of attribute closure algorithm to show there is only one tuple in the join result for each tuple in R_i

Correctness of 3NF Decomposition Algorithm (Cont.)

- Claim: if a relation R_i is in the decomposition generated by the above algorithm, then R_i satisfies 3NF.
- Proof:
  - Let R_i be generated from the dependency $\alpha \rightarrow \beta$
  - Let $\gamma \rightarrow B$ be any non-trivial functional dependency on R_i. (We need only consider FDs whose right-hand side is a single attribute.)
  - Now, B can be in either β or α but not in both. Consider each case separately.

Correctness of 3NF Decomposition (Cont.)

- Case 1: If B in β:
  - If γ is a superkey, the 2nd condition of 3NF is satisfied
  - Otherwise α must contain some attribute not in γ
  - Since $\gamma \rightarrow B$ is in F^+ it must be derivable from F_c, by using attribute closure on γ.
  - Attribute closure not have used $\alpha \rightarrow \beta$. If it had been used, α must be contained in the attribute closure of γ, which is not possible, since we assumed γ is not a superkey.
  - Now, using $\alpha \rightarrow (\beta - \{B\})$ and $\gamma \rightarrow B$, we can derive $\alpha \rightarrow B$ (since $\gamma \subseteq \alpha\ \beta$, and $B \notin \gamma$ since $\gamma \rightarrow B$ is non-trivial)
  - Then, B is extraneous in the right-hand side of $\alpha \rightarrow \beta$; which is not possible since $\alpha \rightarrow \beta$ is in F_c.
  - Thus, if B is in β then γ must be a superkey, and the second condition of 3NF must be satisfied.

Correctness of 3NF Decomposition (Cont.)

- Case 2: B is in α.
  - Since α is a candidate key, the third alternative in the definition of 3NF is trivially satisfied.
  - In fact, we cannot show that γ is a superkey.
  - This shows exactly why the third alternative is present in the definition of 3NF.

Q.E.D.

First Normal Form

- Domain is **atomic** if its elements are considered to be indivisible units
  - Examples of non-atomic domains:
    - Set of names, composite attributes
    - Identification numbers like CS101 that can be broken up into parts
- A relational schema R is in **first normal form** if the domains of all attributes of R are atomic
- Non-atomic values complicate storage and encourage redundant (repeated) storage of data
  - Example: Set of accounts stored with each customer, and set of owners stored with each account
  - We assume all relations are in first normal form (and revisit this in Chapter 22: Object Based Databases)

First Normal Form (Cont.)

- Atomicity is actually a property of how the elements of the domain are used.
  - Example: Strings would normally be considered indivisible
  - Suppose that students are given roll numbers which are strings of the form *CS0012* or *EE1127*
  - If the first two characters are extracted to find the department, the domain of roll numbers is not atomic.
  - Doing so is a bad idea: leads to encoding of information in application program rather than in the database.